PHYS 1301 Handout 4: Data, units and tables

Read the procedure before the lab session. Record every measurement with its unit and an estimate of its uncertainty, and keep your raw data even when a trial looks wrong. Read the procedure before the lab session. Record every measurement with its unit and an estimate of its uncertainty, and keep your raw data even when a trial looks wrong.

Warm-up

The rotation matrix and the linear system below are used in Problem 3.

$$R = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right]$$

$$\begin{aligned} x(t) &= x_0 + v_0 t + \tfrac{1}{2} a t^2 \\ v(t) &= v_0 + a t \\ v^2 &= v_0^2 + 2a(x - x_0) \end{aligned}$$

$$f(x) = \begin{cases} 0 & x < 0 \\ x^2 & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

1. Which quantity is a vector?

(a) mass

(b) speed

(c) displacement

(d) energy

Table 1: Cart on an incline: measured times

Trial	Mass (kg)	Time (s)	Distance (m)	Speed (m/s)
1	0.30	1.27	1.50	1.181
2	0.35	1.34	1.50	1.119
3	0.40	1.41	1.50	1.064
4	0.45	1.48	1.50	1.014
5	0.50	1.55	1.50	0.968
6	0.55	1.62	1.50	0.926
7	0.60	1.69	1.50	0.888
8	0.65	1.76	1.50	0.852
9	0.70	1.83	1.50	0.820
10	0.75	1.90	1.50	0.789

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Table 2: Physical constants used in this course

Quantity	Symbol	Value	Unit
Speed of light	c	2.998×10^{8}	m/s
Gravitational constant	G	6.674×10^{-11}	N m^2/kg^2
Planck constant	h	6.626×10^{-34}	J s
Boltzmann constant	k_{B}	1.381×10^{-23}	J/K
Elementary charge	e	1.602×10^{-19}	C
Avogadro constant	N_{A}	6.022×10^{23}	1/mol

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Table 3: Thermal properties of selected materials

Material	Density (kg/m^3)	Specific heat (J/(kg K))		Melting point (°C)
		Solid	Liquid	
Aluminium	2700	900	1180	660
Copper	8960	385	530	1085
Lead	11340	128	140	327
Water (ice)	917	2090	4186	0

Table 4: Lab report grading rubric

Criterion	Excellent (4)	Good (3)	Needs work (1)
Hypothesis	Clear, testable and linked to the theory in the pre-lab.	Testable but not linked to theory.	Missing or not testable.
Data	All measurements recorded with units and uncertainties.	Units present; some uncertainties missing.	Units or values missing.
Analysis	Correct graph with best-fit line and slope uncertainty.	Graph correct; uncertainty not propagated.	No graph or wrong variables.
Conclusion	Answers the question and discusses sources of error.	Answers the question only.	Does not answer the question.

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Table 5: Course schedule

Week	Dates	Topic	Reading	Due
1	Jan 13-17	Units and measurement	Ch. 1	-
2	Jan 20-24	Motion in one dimension	Ch. 2	HW 1
3	Jan 27-31	Vectors and projectiles	Ch. 3	HW 2
4	Feb 3-7	Newton's laws	Ch. 4	Lab 1
5	Feb 10-14	Friction and circular motion	Ch. 5	HW 3
6	Feb 17-21	Work and energy	Ch. 6	Quiz 1
7	Feb 24-28	Momentum	Ch. 7	HW 4
8	Mar 3-7	Midterm review	-	Midterm

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Table 6: Values of the trigonometric functions

Angle (deg)	sin	cos	tan
0	0.0000	1.0000	0.0000
2	0.0349	0.9994	0.0349
4	0.0698	0.9976	0.0699
6	0.1045	0.9945	0.1051
8	0.1392	0.9903	0.1405
10	0.1736	0.9848	0.1763
12	0.2079	0.9781	0.2126
14	0.2419	0.9703	0.2493
16	0.2756	0.9613	0.2867
18	0.3090	0.9511	0.3249
20	0.3420	0.9397	0.3640
22	0.3746	0.9272	0.4040
24	0.4067	0.9135	0.4452
26	0.4384	0.8988	0.4877
28	0.4695	0.8829	0.5317
30	0.5000	0.8660	0.5774
32	0.5299	0.8480	0.6249
34	0.5592	0.8290	0.6745
36	0.5878	0.8090	0.7265
38	0.6157	0.7880	0.7813
40	0.6428	0.7660	0.8391
42	0.6691	0.7431	0.9004
44	0.6947	0.7193	0.9657
46	0.7193	0.6947	1.0355
48	0.7431	0.6691	1.1106
50	0.7660	0.6428	1.1918
52	0.7880	0.6157	1.2799
54	0.8090	0.5878	1.3764
56	0.8290	0.5592	1.4826

Angle (deg)	sin	cos	tan
58	0.8480	0.5299	1.6003
60	0.8660	0.5000	1.7321
62	0.8829	0.4695	1.8807
64	0.8988	0.4384	2.0503
66	0.9135	0.4067	2.2460
68	0.9272	0.3746	2.4751
70	0.9397	0.3420	2.7475
72	0.9511	0.3090	3.0777
74	0.9613	0.2756	3.4874
76	0.9703	0.2419	4.0108
78	0.9781	0.2079	4.7046
80	0.9848	0.1736	5.6713
82	0.9903	0.1392	7.1154
84	0.9945	0.1045	9.5144
86	0.9976	0.0698	14.3007
88	0.9994	0.0349	28.6363
90	1.0000	0.0000	undefined

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Table 7: Data for the inner planets

	Mercury	Venus	Earth	Mars
Mass (10^{24} kg)	0.330	4.87	5.97	0.642
Radius (km)	2440	6052	6371	3390
Orbital period (days)	88.0	224.7	365.2	687.0
Surface gravity (m/s^2)	3.7	8.9	9.8	3.7

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Instructor	Dr. Alex Rivera
Office	Science Building 2.104
Office hours	Tuesday and Thursday, 2:00-3:30 pm
Email	arivera@example.edu

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Table 8: Labs by unit

Unit	Topic	Lab activity
Mechanics	Kinematics	Photogate timing of a falling ball
	Forces	Atwood machine
	Energy	Spring constant from oscillations
Waves	Sound	Resonance tube
	Light	Diffraction grating

Table 9: Quiz scores of 40 students

Score interval	Frequency	Relative frequency	Cumulative frequency
50-59	3	0.075	3
60-69	7	0.175	10
70-79	14	0.350	24
80-89	11	0.275	35
90-100	5	0.125	40
Total	40	1.000	

Problems

1. Plot distance against time squared for Table 1 and find the acceleration.
2. Use Table 2 to compute the gravitational force between two 1.0 kg masses 1.0 m apart.
3. Solve the linear system in the warm-up and apply R with $\theta = 30^\circ$ to the solution.